Tobacco Use & COPD Tracker: A Shiny-Based Digital Health Application

Module GPH-01: Digital Health

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30.06.2026

1. Introduction and Objectives

Digital health tools play an increasingly important role in supporting evidence-based decision-making by making complex public health data accessible and interpretable. The objective of this project was to develop a simple, interactive Shiny application that visualizes the relationship between tobacco use prevalence and COPD prevalence across three countries, broken down by sex and tracked over three time points.

Rather than aiming for clinical prediction or individual risk assessment, the application focuses on supporting public health authorities in identifying patterns at the country level. The goal is to make it easier to see where the burden is concentrated, which in turn can help guide where attention and resources are most needed.

2. Context and Background

Tobacco use is the leading preventable cause of COPD worldwide. The relationship between the two is well established in the literature, yet the way this relationship plays out differs considerably between countries. Differences in smoking history, age structure, healthcare access, and data collection methods all contribute to variation in how tobacco prevalence and COPD burden align across settings.

From a public health perspective, comparing countries on these two indicators side by side over time offers a useful lens for prioritizing interventions and monitoring progress. This project reflects that perspective by presenting both indicators together in a single view, allowing users to observe their relationship directly rather than consulting separate databases.

3. Key User Persona

The primary user group of the application consists of public health professionals, researchers, and health policy analysts who want a quick overview of tobacco and COPD trends across countries, without needing to download and combine data from multiple sources manually.

Key characteristics of this user group include:

- Familiarity with public health concepts but not necessarily with data tools
- Need for quick, comparable cross-country figures
- Interest in trends over time rather than a single snapshot

As a result, the application prioritizes simplicity and directness over analytical depth. The outputs are limited to what is most immediately useful for a first-pass comparison.

4. Application Design

The application uses three filter inputs:

- Country (United States, United Kingdom, Germany)
- Sex (Male, Female)

- Year (2000, 2012, 2022)

These filters control two headline figures displayed at the top of the results panel, showing tobacco prevalence and COPD prevalence for the selected combination. Below these, a bar chart compares tobacco prevalence across all three countries for the selected year and sex group, with the COPD rate for the selected country shown as a dashed horizontal line.

This design allows users to see not only the absolute values for a selected country but also how that country sits relative to the others. The dashed COPD line adds a second dimension to the chart without introducing a separate visualization, keeping the interface compact.

5. Visualization Types

The application includes one visualization:

Bar Chart with Overlaid Reference Line: A bar chart displays tobacco use prevalence for all three countries under the selected filter conditions, with the selected country highlighted in a distinct color. A dashed horizontal line marks the COPD prevalence for the selected country, allowing users to read both indicators from the same chart.

This approach was chosen over separate charts for tobacco and COPD because it makes the relationship between the two indicators visible directly. Users do not need to compare numbers across two separate plots; they can see how the smoking burden of a country relates to its COPD rate in a single view.

6. Implementation Challenges

The most significant limitation of the current version is country coverage. The application covers only three countries: the United States, the United Kingdom, and Germany. This was a deliberate decision driven by data quality rather than a constraint of the tool itself.

Both the WHO and the IHME publish tobacco and COPD figures for a much wider range of countries. However, for many lower- and middle-income countries, the underlying data relies on older surveys, sparse reporting, or model-based projections with considerable uncertainty. Presenting these figures alongside higher-quality estimates without distinguishing between them would risk misleading users. Expanding country coverage in a responsible way requires either limiting the selection to settings with reliable data or building in a way to communicate data quality alongside the estimates, which was outside the scope of this version.

Broader country coverage remains the most meaningful next step for this application, as the tool is most useful precisely in settings where the tobacco-COPD relationship is less well documented and where data-driven prioritization could have the greatest impact.

Disclaimer

This application and accompanying report are for educational purposes only and do not provide clinical or policy recommendations. Data is sourced from the WHO Global Health Observatory and the IHME Global Burden of Disease study.

App Link:

https://github.com/mostafagad-emd/tobacco-copd-tracker/blob/main/app_tobacco_copd_v2%20(3).R